Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

Tick Mark (✓) the correct answer

1. The branch of Chemistry which deals with hydrocarbons:

   (a) Industrial chemistry (b) Inorganic chemistry

   (c) Organic chemistry (d) Physical chemistry

2. The atomic mass of an element expressed in gram is:

   (a) Gram molecular mass (b) Gram atomic mass

   (c) Gram formula mass (d) Mole

3. Which of the following can be separated by physical means?

   (a) Mixture (b) Element

   (c) Compound (d) Substance

4 The molar mass of H_2SO_4 is:

   (a) 98 a.m.u (b) 9.8gm

   (c) 98gm (d) 9.8 a.m.u

5. The Molecule consist of two atoms is :

   (a) Monoatomic molecule (b) Polyatomic molecule

   (c) Hetero atomic molecule (d) Di atomic molecule

6. A formula that indicates actual number and type of atoms in a molecule is called:

   (a) Chemical formula (b) Empirical formula

   (c) Molecular formula (d) Formula mass

7. Ethyl alcohol was prepared by:

   (a) Ibne-Sina (b) Al-Razi

   (c) Al-Beruni (d) Jaber bin-Hayan

8. which of the following is a homo atomic molecule:

   (a) H_2 (b) NH_3

   (c) H_2O (d) CO_2

9. The Empirical formula of hydrogen peroxide (H_2O_2) is:

   (a) H_2O_2 (b) HO

   (c) OH_2 (d) O_2H_2

10. A piece of matter in pure form is termed as:

   (a) Redical (b) Mixture

   (c) Compound (d) substance

SECTION- B: SHORT QUESTIONS:

1. Differentiate between the physical and analytical chemistry?
2. Write down the classification of molecule?
3. Identify the differences among the following:
   (a) Atom and Ion
   (b) Molecule and Molecular ion
   (c) Compound and Mixture
4. Define the following terms:
   (a) Gram atomic mass
   (b) Gram molecular mass
   (c) Gram formula mass
5. Write down the empirical and molecular formula of the following?
   Sulphuric acid, Carbon dioxide, Glucose, Benzene
6. What is Free Radical?
7. Describe relationship between empirical and molecular formula? Explain with examples.
8. Explain why hydrogen and oxygen are considered as element whereas water is compound?

SECTION- C: DETAILED QUESTIONS:

1. What do you mean chemical species, explain ion, molecular ion and free radical?
2. Write down the applications of chemistry in daily life?
3. Explain in detail empirical and molecular formula?
4. Explain the steps for balancing the equation?
5. Name the branches of chemistry and discuss any five branches?

EXERCISE

SECTION- A: MULTIPLE CHOICE QUESTIONS

Tick Mark (✓) the correct answer

1. In an atom number of protons and neutrons are added to obtain:
   (a) number of electrons (b) number of nucleons
   (c) atomic number of element (d) number of isotopes
2. If proton number is 19, electron configuration will be:
   (a) 2, 8, 9 (b) 2, 8, 8, 1
   (c) 2, 8, 1 (d) 2, 8, 3
3. If nucleon number of potassium is 39 and its atomic number is 19 then, number of neutrons will be:
   (a) 39 (b) 19
   (c) 20 (d) 29
4. The isotope C-12 is present in abundance of:
   (a) 96.9% (b) 97.6%
   © 98.89% (d) 99.7%
5. Electronic configuration is distribution of:
   (a) protons (b) neutrons
   (c) electrons (d) positrons
6. Which one of the following is most penetrating?
   (a) electron (b) Proton
   (c) alpha particle (d) neutron
7. How many subshells in a L shell:
   (a) one (b) two
   (c) three (d) four
8. De Broglie extend the wave particle duality to electron in:
   (a) 1920 (b) 1922
   (c) 1923 (d) 1925
9. Name the material of screen which is used in Rutherford atomic model :
   (a) Aluminum foil (b) zinc sulphide
   (c) sodium sulphide (d) Aluminum sulphide
10. Which rays are used for sterilization of medical instruments :
    (a) α-rays (b) β-rays
    (c) γ-rays (d) x-rays



Summary

- 19th century is considered as milestone in Systematic arrangment of elements in Periodic Table.
- Dobereiner arranged elements in Triads.
- Newland put forward Law of Octaves.
- Mendeleev Published Preiodic law with groups and Rows.
- Moseley stated his law as "The physical and chemical properties of elements are the periodic function of their atomic numbers"
- There are total eighteen groups and seven periods in modern periodic table.
- Physical and Chemical properties change from left to right in a period. Elements of a period show different properties because the electronic configuration continuously changes within a period.
- The sub groups are divided on the basis of their similar properties as A and B and placed together in periodic table.
- The elements of sub group A are called Main or Representative Elements.
- The elements of sub group B are called Transition Elements. The group number indicate total number of electrons in valence shell of the element.
- The atomic size increases down a group but decreases along the period.
- Ionization energy decreases down a group but increases along a period.
- Electronegativity decreases down a group but increases along a period.
- Electron affinity decreases down a group but increases along a period.
- The Shielding effect increases down the group in periodic table and remain same in period from left to right.

EXERCISE

SECTION- A: MULTIPLE CHOICE QUESTIONS

Tick Mark (✓) the correct answer

1. In 1869 Mandeleev put forward his periodic law about:

   (a) Atomic Number (b) Chemical properties

   (c) Physical properties (d) Atomic Mass

2. The periodic table divided into s, p, d, and f block based on.

   (a) Atomic Radius (b) Electronic Configuration

   (c) Ionization Energy (d) Electron Affinity

3. 4th and 5th period in periodic table are known as:

(a) Short period (b) Long period

(c) Normal period (d) Very long period

4. Which one of the following decreases along the period?

(a) Ionization Energy (b) Atomic Radius

(c) Electronegativity (d) Electron Affinity

5. The elements of VIIA group are known as:

(a) Lanthanides (b) Actinides

(c) Halogens (d) Nobel Gases

6. According to Mosely the chemical properties of elements are the periodic function of their :

(a) Atomic Size (b) Atomic Mass

(c) Atomic Radius (d) Atomic Number

7. The shielding effect across the period :

(a) Increases (b) Decrease

(c) Moderate (d) Same

8. The ability to attract shared pair of electron is called:

(a) Electron Affinity (b) Electronegativity

(c) Ionization Energy (d) Shielding Effect

9. In group electron affinity values decreases from top to bottom because :

(a) Atomic size normal (b) Atomic size increases

(c) Atomic size decreases (d) Atomic size same

10. All Transition Elements are :

(a) Gases (b) Metals

(c) Nonmetals (d) Metalloids

EXERCISE

SECTION- A: MULTIPLE CHOICE QUESTIONS

Tick Mark (✓) the correct answer

1. An example of ionic compound is:

   (a) H_2 (b) CH_4 (c) N_2 (d) NaCl

2. Interaction between highly electron deficient hydrogen and highly electronegative atom is called

   (a) covalent bond (b) ionic bond

   (c) hydrogen bond (d) metallic bond

3. Two fluorine atoms share one electron each in their outermost shell to achieve electronic configuration of:

   (a) Xe (b) Ar

   (c) Kr (d) Ne

4. Number of electrons lost by atoms of group IIIA equals:

   (a) 1 (b) 2

   (c) 3 (d) 4

5. Atom which loses two electrons from its outer shell to form ion is:

   (a) oxygen (b) potassium

   (c) magnesium (d) carbon

6. In NaCl crystal lattice each Na^+ ion is surrounded by:

   (a) 6 Cl^- ions (b) 6 Na^+ ions

   (c) 8 Cl^- ions (d) 12 Cl^- ions

7. At room temperature most of ionic compounds are:

   (a) amorphous solids (b) crystalline solids

   (c) liquids (d) gases

8. Tendency of atoms to acquire eight electrons in their valence shell is:

   (a) octet rule (b) duplet rule

   (c) triplet rule (d) none of above

9. When one atom forms cation by losing electron and other forms anion by accepting that electron then bond form between them is:

   (a) Covalent bond (b) Ionic bond

   (c) coordinate covalent bond (d) hydrogen bond

10. Nobel gases are stable because they contain:
    (a) 4 electrons in valence shell (b) 6 electrons in valence shell
    (c) 8 electrons in valence shell (d) 10 electrons in valence shell
11. Bond which involve 3 shared electron pairs is a :
    (a) double covalent bond (b) single covalent bond
    (b) triple covalent bond (d) none of above
12. A non-metal atom form anion by
    (a) loses of electrons (b) gain of electrons
    (c) loses of protons (d) gain of protons
13. When two identical atoms share electron pairs and exert same force on each other than
    bond form is:
    (a) non-polar covalent bond (b) polar covalent bond
    (c) double covalent bond (d) coordinate covalent bond
14. Synthetic resins are used on places where:
    (a) electric resistance is required (b) water resistance is required
    (c) adhesion is required (d) friction is required
15. Oxygen belongs to group VIA so number of electrons in its valence shell are:
    (a) 4 (b) 2
    (c) 6 (d) 8
16. Electron pairs which are not shared by atoms are called:
    (a) electron pairs (b) lone pairs
    (c) bond pairs (d) shared pairs
17. Strength of intermolecular forces from ionic or covalent bond is:
    (a) Weaker (b) stronger
    (c) equal (d) none of above
18. Ionic crystals have:
    (a) high melting points (b) moderate melting points
    (c) low melting points (d) none of above
19. Bond formed by mutual sharing of electron is:
    (a) ionic bond (b) coordinate covalent bond
    (c) covalent bond (d) metallic bond

20. Which of the following diagram shows atoms are bonded with same electro negativity?

(a) A — B (b) A — B

(c) A — B (d) A — B

SECTION- B: SHORT QUESTIONS:

1. Draw dot and cross diagrams to show how different types of chemical bonds are formed when fluorine reacts with
   (a) Hydrogen (b) potassium
2. What is meant by octet and duplet rule?
3. Can you draw an ion which is formed by the atom losing three electrons?
4. How oxygen forms an anion?
5. What is the difference between lone pair and bond pair?
6. Explain why table salt has a very high melting point.
7. How is electronegative value determined the formation of chemical bond?
8. Why it is easy for magnesium atom to lose two electrons?
9. Atoms of metallic elements can form ionic bond, but they are not very good to form covalent bonds. Why?
10. How does an ion differ from an atom?
11. Describe dipole-dipole forces.
12. Write uses of adhesive material.
13. Why Intermolecular forces are weaker than intera molecular forces?
14. Write characteristics of metallic bond.
15. Ionic bonds are strong and hard to break but why most of the covalent compounds have low melting and boiling points.
16. Write down the characteristics of ionic compounds.
17. Why ionic compounds are solid?
18. How is hydrogen bonding affecting the physical properties of compounds?
19. Complete the chart:

Atomic Number	Number of protons	Number of electrons	Electronic configuration	Number of valence electrons
11	11	11	2, 8, 1	1
12				
13				
14				
15				
16				

Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

Tick Mark (✓) the correct answer

1. Which of the following gases diffuse fastest?
   (a) Hydrogen (b) Chlorine
   (c) Fluorine (d) Helium
2. The vapour pressure of a liquid increases with the:
   (a) increase of pressure
   (b) increase of temperature
   (c) increase of intermolecular forces
   (d) increase the polarity of molecules
3. The freezing point depends upon:
   (a) Nature of liquid (b) Pressure
   (c) Temperature (d) Volume
4. One atmospheric pressure is equal to:
   (a) 10325 Pascal (b) 106075 Pascal
   (c) 10523 Pascal (d) 101325 Pascal
5. Which of the following does not affect the boiling point :
   (a) Intermolecular forces (b) External pressure
   (c) Initial temperature of liquid (d) Nature of liquid
6. The mobility of liquids is lesser than:
   (a) Solids (b) Gases
   (c) Both a & b (d) None of these
7. Which of the following have sharp melting point in solids :
   (a) Plastic (b) Rubber
   (c) Glass (d) Diamond
8. Which of the following is the lightest form of matter:
   (a) Solid (b) Liquid
   (c) Gases (d) Plasma
9. The liquid molecules leave the surface of liquid in evaporation process because :
   (a) Energy is low (b) Energy is moderate
   (c) Energy is high (d) None of these
10. The density of gases increases when its :
   (a) Pressure increased (b) Temperature increased
   (c) Volume increased (d) None of these

EXERCISE

SECTION- A: MULTIPLE CHOICE QUESTIONS

Tick Mark (✓) the correct answer

1. An alloy is the homogeneous mixture of:
   (a) two solids (b) two liquids
   (c) two gases (d) solid and liquid
2. A saturated solution of KCl on heating becomes:
   (a) unsaturated (b) supersaturated
   (c) diluted (d) all of these
3. If we dissolve sand into the water, then the mixture is said to be:
   (a) solution (b) suspension
   (c) colloids (d) concentrated solution
4. Solubility is usually expressed in grams of the solute dissolved in _____grams of a solvent
   (a) 10 (b) 100
   (b) 500 (d) 1000
5. Example of heterogeneous mixture is:
   (a) sugar and water (b) sand and water
   (c) salt and water (d) ink and water
6. 2 moles of sodium chloride (NaCl) is equal to:
   (a) 123 grams (b) 135 grams
   (c) 158 grams (d) 117 grams
7. Molarity of a solution which is prepared by dissolving 40g sodium chloride in 500cm^3 of solution is:
   (a) 1.4M (b) 1.5M
   (c) 1.6M (d) 1.7M
8. 10% (w/w) sugar solution mean that 10 grams of solute dissolved in:
   (a) 90g of water (b) 95g of water
   (c) 100g of water (d) 105g of water

9. An example of true solution is:
   (a) solution of starch (b) solution of soap
   (c) ink in water (d) tooth paste
10. Which solution contain more water:
   (a) 1.0M (b) 0.75M
   (c) 0.5M (d) 0.25M
11. When a saturated solution is diluted, it change into:
   (a) saturated solution (b) unsaturated solution
   (c) concentrated solution (d) supersaturated solution
12. Butter is example of solution:
   (a) gas-liquid (b) solid-liquid
   (c) liquid-solid (d) gas-solid
13. A solution that contain solid solute into liquid solvent is called:
   (a) solids in gas (b) liquids in solids
   (c) solids in solids (d) solids in liquid
14. What is the particle size in suspension?
   (a) 10^3nm (b) 10^2 nm
   (c) less than 10^3nm (d) greater than 10^3nm
15. Write the example of each type of solution.

Solute	Solvent	Example
Solid	Liquid	
Gas	Gas	
Solid	Solid	
Liquid	Solid	
Liquid	Gas	
Liquid	Liquid	

Exercise

SECTION- A: MULTIPLE CHOICE QUESTIONS

Tick Mark (✓) the correct answer

1. Alloy of Cu - Sn is called _____
   (a) Brass (b) Bronze
   (c) Monel (d) Bell metal
2. Which one is Alloy.
   (a) Graphite (b) Mercury
   (c) Steel (d) Water
3. Value of 1 Farady is equal to _____.
   (a) 9.65 C (b) 9650 C
   (c) 96500 C (d) 965 C
4. Which one is non electrolyte.
   (a) Aqueous HCl (b) Aqueous NaCl
   (c) Molten KCl (d) Urea
5. Which one is oxidizing agent.
   (a) Al (b) H_2S
   (c) Cl_2 (d) NaH
6. Which one is reducing agent.
   (a) H_2SO_4 (b) HNO_3
   (c) Al (d) I_2
7. Which one forms weak electrolyte solution with water
   (a) HCl (b) KOH
   (c) NaCl (d) CH_3COOH
8. In dry cell _____ is used as cathode.
   (a) Zn (b) Cu
   (c) Sn (d) graphite
9. 1g equivalent weight of Al is equal to _____.
   (a) 9 g (b) 27g
   (c) 54 g (d) 1 g

10. Which one is correct statement.
    (a) Oxidation occurs at cathode.
    (b) Reduction occurs at anode
    (c) Reduction occurs at cathode
    (d) Ions lose electrons at cathode.

SECTION- B: SHORT QUESTIONS:

1. Define oxidation, reduction reactions with examples.
2. Why ionic compounds conduct electricity in molten or in aqueous solutions only?
3. What is electrolytic cell? Explain with diagram.
4. Define oxidizing and reducing agent with examples.
5. Examine the following chemical equations and identify.
   (i) Oxidizing agent (ii) Reducing agent
   (iii) Substance undergoes oxidation
   (iv) Substance undergoes reduction

   1. $Zn + Cl_2 \longrightarrow ZnCl_2$
   2. $Br_2 + H_2S \longrightarrow 2HBr + S$
   3. $2Ca + O_2 \longrightarrow 2CaO$
   4. $2Li + S \longrightarrow Li_2S$

Identify the Alloy.

Components	Alloy
Cu – Zn	
Cu – Al – Mg – Ni	
Cu – Zn – Sn	

SECTION- C: DETAILED QUESTIONS:

1. Describe the dry cell with diagram.
2. What is battery? How lead storage battery works?
3. Explain process of electrolysis in electrolytic cell.
4. What is Alloy? Explain its classification with examples.
5. What is rusting? How it can be prevented.
6. What is electroplating? How steel object can be electroplated with Tin, Zinc and Silver?
7. State and explain Faraday first and second law of electrolysis.

Exercise Questions

SECTION- A: MULTIPLE CHOICE QUESTIONS

Tick Mark (✓) the correct answer

1. Which one metal belongs to Alkaline earth metal?
   (a) B (b) Bi (c) Br (d) Ba
2. Which one is Barium
   (a) Bi (b) Be (c) Ba (d) Br
3. Chlorine can be displaced by _____ .
   (a) F (b) Br (c) I (d) At
4. Which one is strong acid?
   (a) HF (b) HCl (c) HBr (d) HI
5. Which Halogen exists in liquid state?
   (a) F_2 (b Cl_2 (c) Br_2 (d) I_2
6. Non-metals of _____ group are gases
   (a) VI-A (b) VII-A (c) VIII-A (d) VIII-B
7. Which one is Metalloid?
   (a) Br (b) Se (c) S (d) Sr
8. Which one of the following act as oxidizing agent ?
   (a) Be (b) Mg (c) Na (d) Cl
9. Which gas can turn lime water milky?
   (a) O_2 (b) NO_2 (c) CO_2 (d) N_2
10. Which compound is known as lunar caustic?
    (a) KNO_3 (b) $AgNO_3$ (c) NaOH (d) $NaNO_3$